

Model A · Diabetic Retinopathy Grading · Linear Probe

Frozen RETFound-DINOv2 ViT-Large backbone · Linear head only · 50 epochs · 4-class (R0/R1/R2/R3A)

What is Linear Probing?

All ~307 M backbone weights are FROZEN — gradients flow only through a single linear layer (1024 → 4). This measures how much retinopathy information is already encoded in the RETFound-DINOv2 representations without any task-specific adaptation, establishing a clean lower bound before the more expensive full fine-tune.

0.837

Best Val AUROC

0.640

Macro Sensitivity

0.864

Macro Specificity

Training Configuration

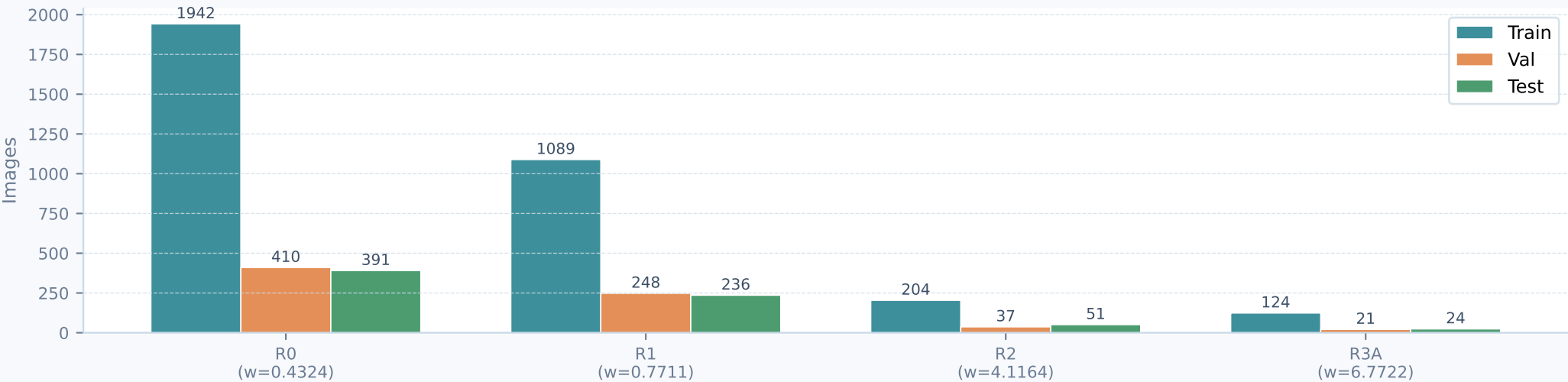
Backbone	RETFound-DINOv2 ViT-Large (~307M params)
Pretrained on	UK Biobank + MEH fundus cohort (736k images)
Backbone status	FROZEN — weights not updated
Head	Linear: 1024 → 4 classes (no hidden layers)
Epochs	50 (best ckpt by val AUROC at epoch 45)
Batch size	64
Input	224 × 224 px colour fundus photo
Optimiser	AdamW (default RETFound settings)
Loss	CrossEntropyLoss with inverse-frequency weights
Class weights	R0=1.00 · R1=1.79 · R2=9.53 · R3A=15.68
Best ckpt metric	Val AUROC → checkpoint-best.pth
Test eval	Deferred (run after fine-tune for direct comparison)

Class Imbalance & Weighting

The dataset is severely skewed — R0 outnumbers R3A by ~16:1. Without correction the model would learn to predict R0 for everything, achieving high accuracy while completely missing sight-threatening proliferative disease.

Weight formula: $w_i = N / (n_classes \times count_i)$
Scaled so min weight = 1.0 → each class contributes equally to the loss regardless of frequency.

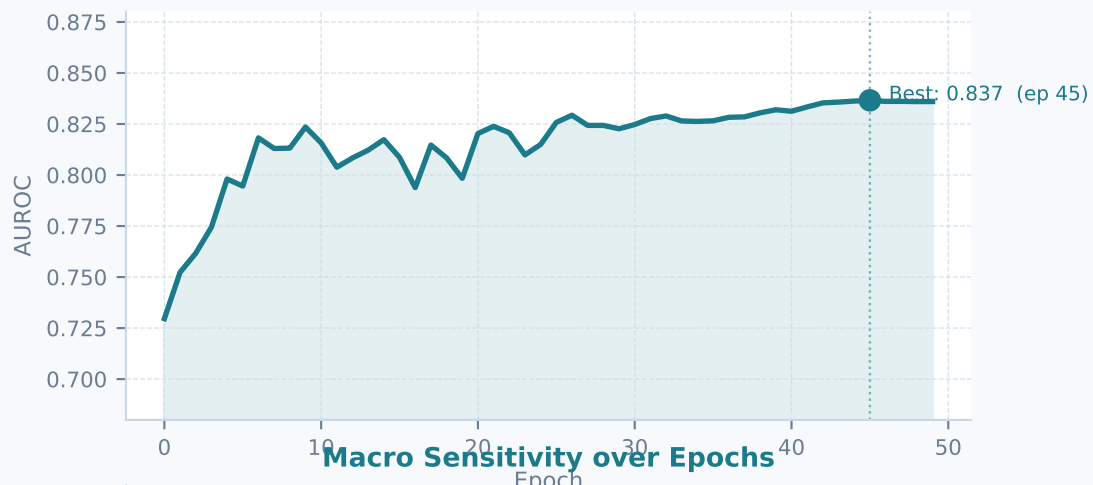
Dataset · Image counts per class and split



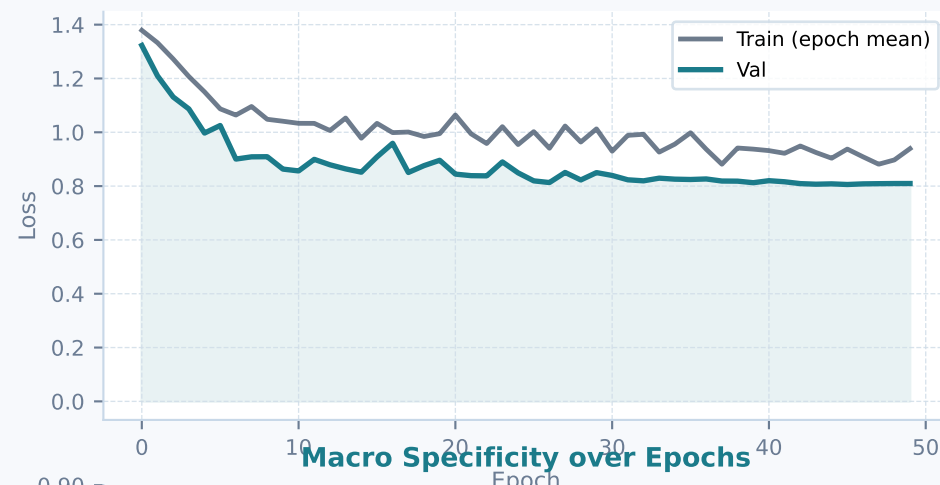
Training Dynamics — All Validation Metrics over 50 Epochs

Frozen backbone · only the linear classification head receives gradient updates each epoch

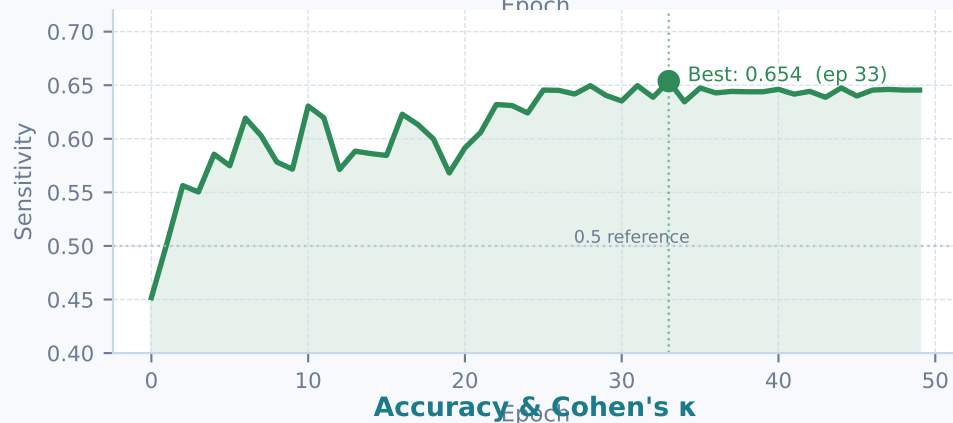
Val AUROC



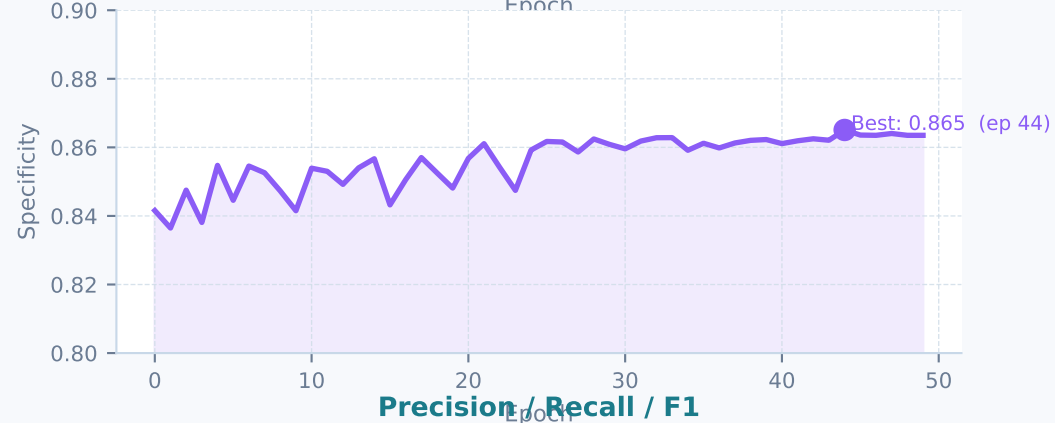
Train Loss vs Val Loss



Macro Sensitivity over Epochs



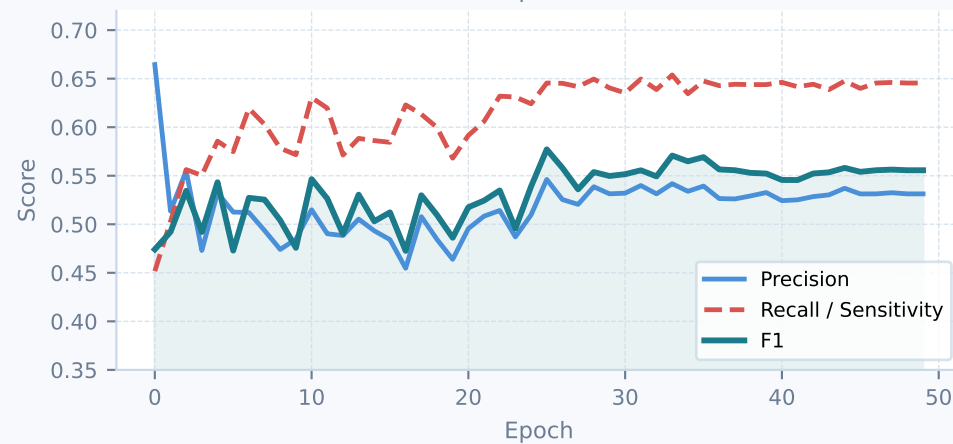
Macro Specificity over Epochs



Accuracy & Cohen's κ



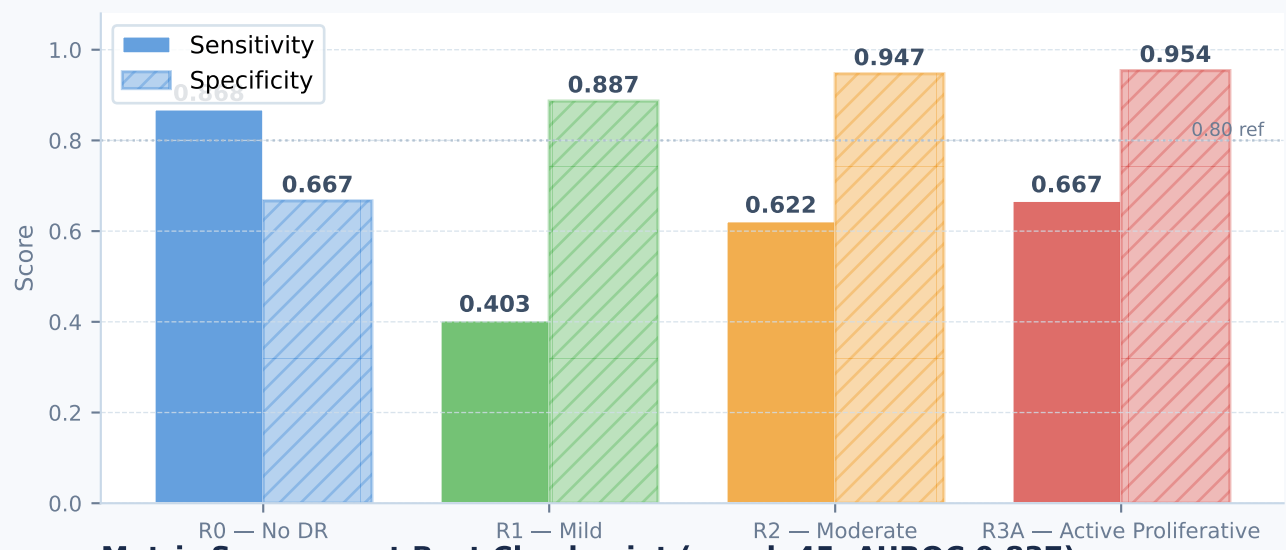
Precision / Recall / F1



Per-class Sensitivity & Specificity — Linear Probe

Sensitivity = TP/(TP+FN) · Specificity = TN/(TN+FP) · OvR decomposition at best-AUROC checkpoint

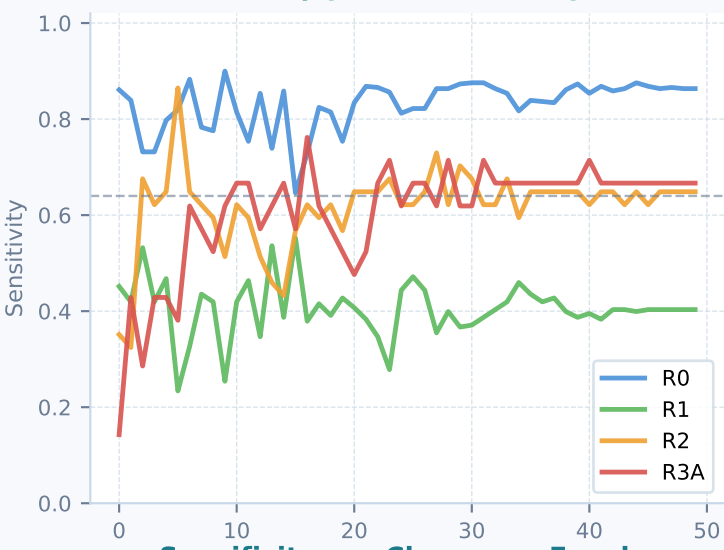
Sensitivity & Specificity per Class (best checkpoint, epoch 45)



Metric Summary at Best Checkpoint (epoch 45, AUROC 0.837)

Class	Sensitivity	Specificity	Clinical Note
R0 — No DR	0.868	0.667	High sensitivity: model catches most healthy eyes
R1 — Mild	0.403	0.887	Weakest sensitivity — mild DR hard to separate from R0 linearly
R2 — Moderate	0.622	0.947	Moderate recall; high specificity avoids over-alerting
R3A — Active Prolif.	0.667	0.954	Good sensitivity despite only 124 training images
Macro average	0.640	0.864	Frozen backbone; full fine-tune expected to improve all classes

Sensitivity per Class over Epochs



Specificity per Class over Epochs



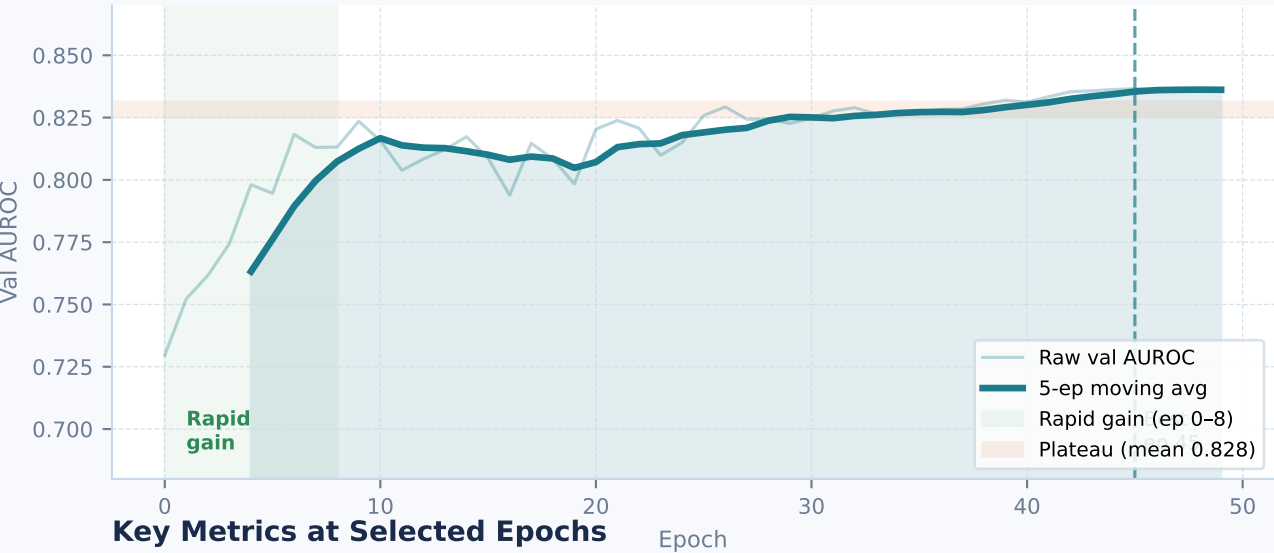
Clinical Interpretation of Sensitivity vs Specificity

- **Sensitivity (recall):** Low sensitivity = missed cases. Clinically dangerous for R2/R3A— undetected proliferative disease can lead to preventable blindness.
- **Specificity:** Low specificity = false alarms. R0 specificity of 0.67 means 33% of healthy eyes are flagged— acceptable as a screening tool but increases unnecessary referrals.
- **R1 bottleneck:** R1 sensitivity of 0.40 is the weakest point. Mild DR is a borderline grade that overlaps with both R0 and R2 in feature space— a linear head cannot form a curved boundary.

Convergence Analysis & Summary

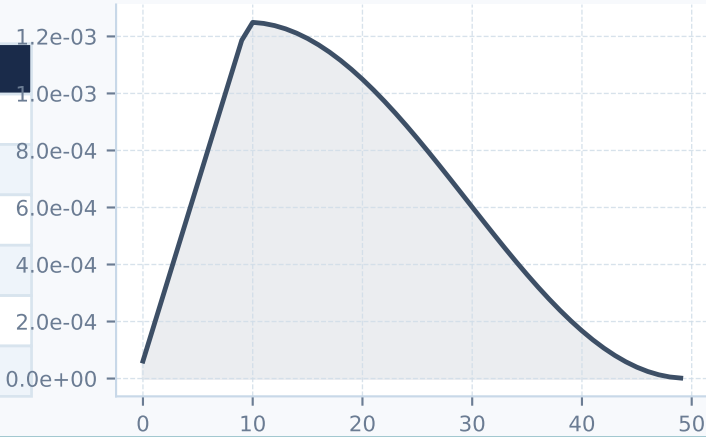
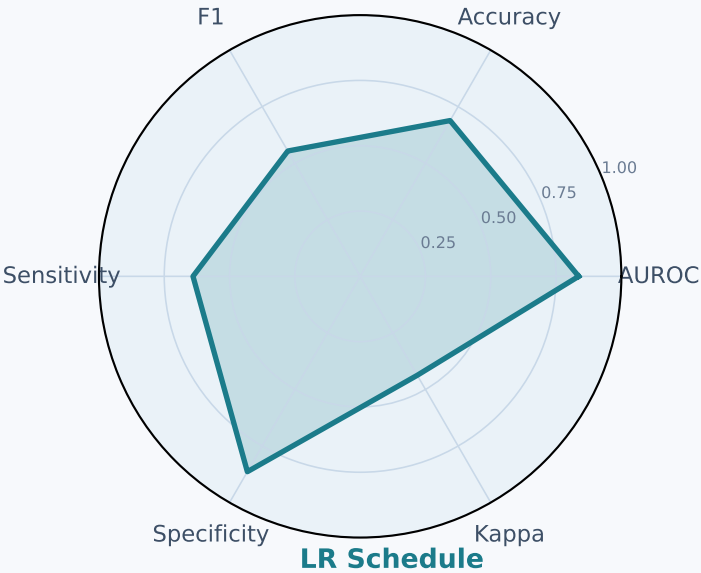
Understanding training behaviour · plateau detection · key takeaways

AUROC Convergence (raw + 5-epoch moving average)



Epoch	AUROC	Accuracy	Kappa	Sensitivity	Specificity	F1
Ep 0	0.7296	0.6718	0.3660	0.4517	0.8416	0.4742
Ep 4	0.7981	0.6648	0.4008	0.5856	0.8547	0.5433
Ep 9	0.8235	0.6480	0.3529	0.5716	0.8415	0.4757
Ep 19	0.7984	0.6243	0.3546	0.5681	0.8481	0.4861
Ep 45 ★	0.8366	0.6885	0.4374	0.6400	0.8636	0.5540
Ep 49	0.8360	0.6872	0.4367	0.6455	0.8635	0.5557

Metric Profile (best epoch 45)



Key Observations

- AUROC climbed rapidly in the first 8 epochs (0.730 → 0.813), confirming RETFound features already carry substantial retinopathy structure without any adaptation.
- The model plateaued around AUROC 0.828 from epoch 20 onwards; the linear head saturated and additional training offered negligible gains.
- Macro specificity (0.864) substantially exceeds macro sensitivity (0.640)— the model is conservative: it rarely raises a false alarm but does miss cases, especially R1.
- R1 sensitivity (0.403) is the critical weakness: mild DR sits in a feature-space region that cannot be cleanly separated from R0 by a single hyperplane.
- R2 and R3A achieve sensitivity of 0.622 / 0.667 respectively despite being rare classes— the class weights force the model to attend to these grades.
- This baseline establishes AUROC 0.837 as the frozen-feature lower bound; full fine-tuning of all backbone layers is expected to push significantly higher.